Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus:	**Bedfortview**
Faculty:	**Information Technology**
Module Code:	**ITDSA2**
Group:	**Group 3**
Lecturer's Name:	**Charity Raphugela**
Student Full Name:	**Olwethu Sithole**
Student Number:	**EDUV6697209**

Indicate	Yes	No
Plagiarism report attached		

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature	Date

Lecturer's Comments:

Marks Awarded:	%

Table of contents

1.1

Entity Name	Associated Attributes
Student	StudentID, FullName, DOB, CellNumber, Address, AdmissionDate, ParentID and SpecializationID
Parent	ParentID, Full name, Pcellphone, Physical address
Specialization	SpecializationID, Title, NQFlevel
Subject	SubjectID, Sname, NumberOfUnits
Exam_Adminstrator	AdministratorID, Ename, Ecellphone, DepartmentID
Campus	CampusID, Cname, Address
Department	DepartmentID, Dname, CampusID
Exam_History	StudentID, ExamID, DateTaken, Mark
Exam	ExamID, Description, DateConducted, Mark, StudentID

1.2

Student:

- StudentID – primary key
- ParentID – foreign key
- SpecializationID – foreign key

Parent:

- ParentID – primary key

Specializations:

- SpecializationID – primary key

Subject:

- SubjectID – primary key
- SpecializationID – primary key

Exam_Administrator:

- AdministratorID – primary key
- DepartmentID – foreign key

Campus:

- CampusID– primary key

Department:

- DepartmentID – primary key
- CampusID – foreign key

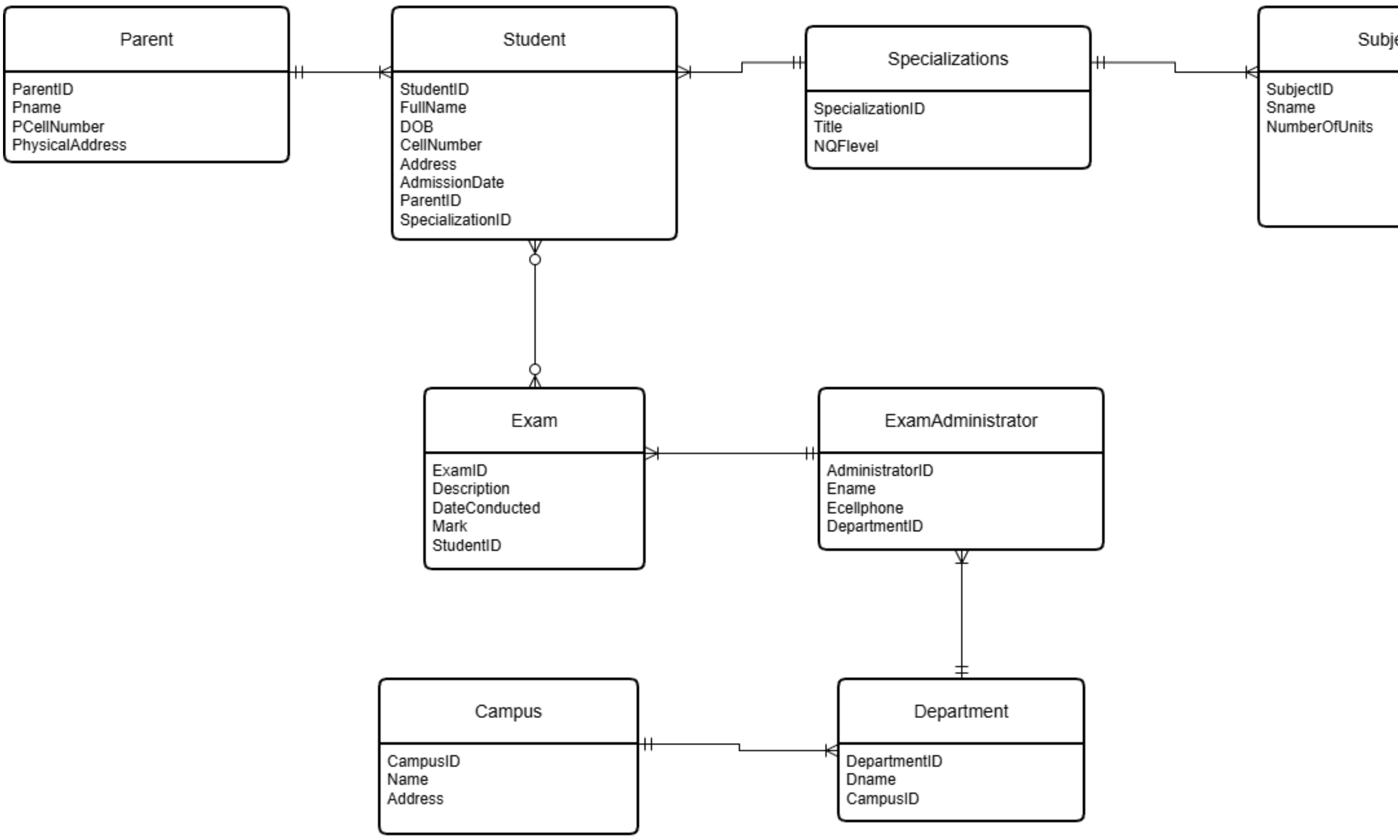
ExamHistory:

- StudentID and ExamID – Composite key
- ExamID – foreign key

Exam:

- ExamID – primary key
- SujectID – foreign key
- AdministratorID – foreign key

1.3



Unresolved many to many between STUDENT and EXAM

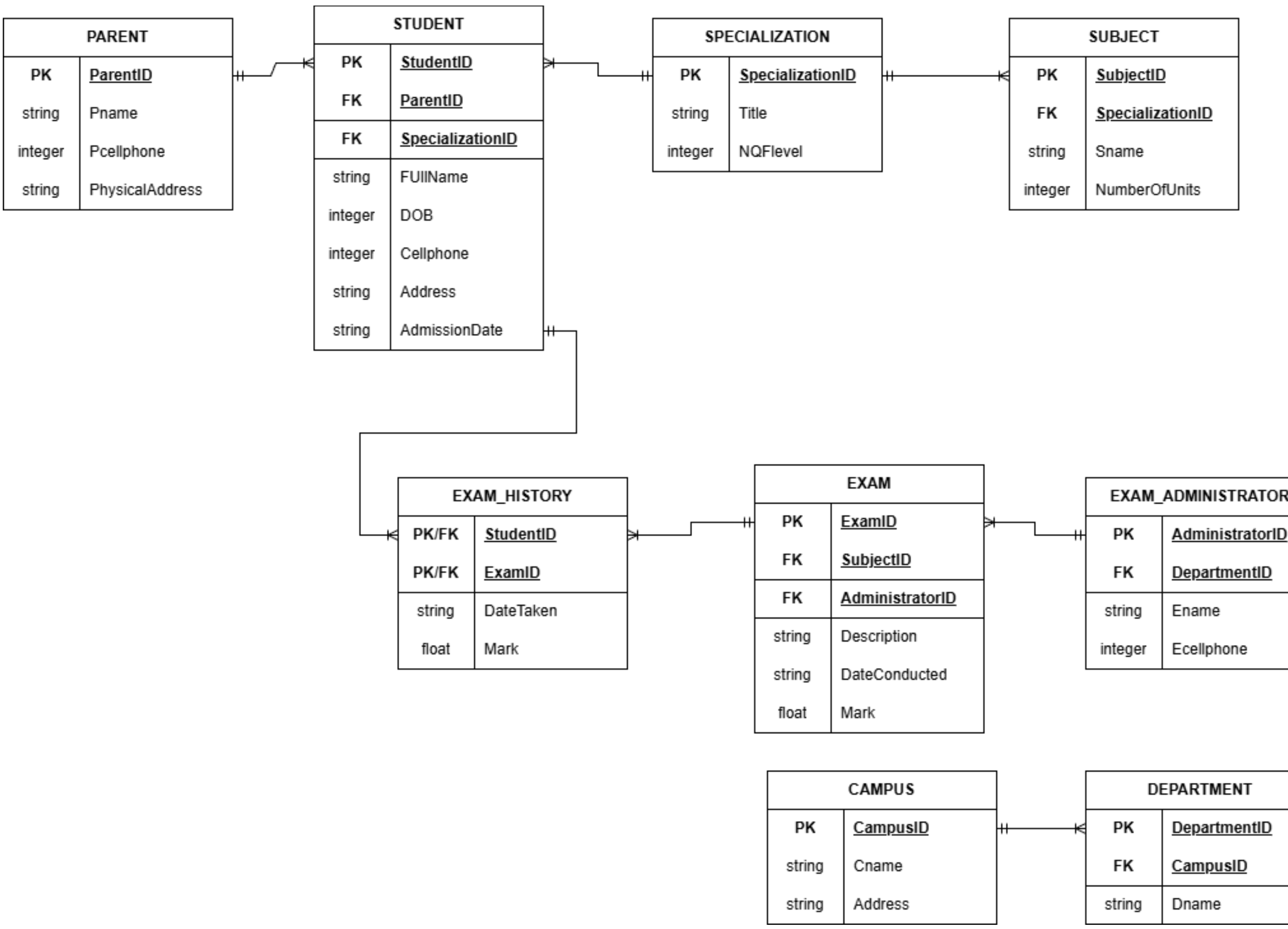
(Ivan Belcic & Cole Stryker, n.d.)

(Coursear Staff, 2026)

(Redgate Data Modeler, 2020)

Question 2

2.1



(*What Is a Logical Data Model?*, n.d.)

2.2

Relationship entity pair	Relationship type
Parent - Student	One to many
Student - Specialization	Many to one
Specialization - Subject	One to many
Student – Exam_history	One to many

Exam_history - Exam	Many to one
Exam – Exam_Administrator	Many to one
Exam_Administrator - Department	Many to one
Department - Campus	Many to one

2.3

Unnormalized form:

STUDENT_INFO:

StudentID
ParentID
SpecializationID
FullName
DOB
Cellphone
Address
AdmissionDate
Pname
Pcellphone
PhysicalAddress
Title
NQFlevel SubjectID
Sname
NumberOfUnits
ExamID
AdministratiorID
Description
DateConducted
Mark
DepartmentID
Ecellphone
Ename
CampusID
Dname
Cname
Address

Normalized form:

PARENT:

ParentID	Pname	Pcellphone	PhysicalAddress

STUDENT:

StudentID	ParentID	SpecializationID	FullName	DOB	Cellphone	Address	AdmissionDate

SPECIALIZATION:

SpecializationID	Title	NQFlevel

SUBJECT:

SubjectID	SpecializationID	Sname	NumberOfUnits

EXAM_HISTORY:

StudentID	ExamID	DateTaken	Mark

EXAM:

ExamID	SubjectID	AdministratiorID	Description	DateConducted	Mark

EXAM_ADMINISTRATOR:

AdministratiorID	DepartmentID	Ename	Ecellphone

DEPARTMENT:

DepartmentID	CampusID	Dname

CAMPUS:

CampusID	Cname	Address

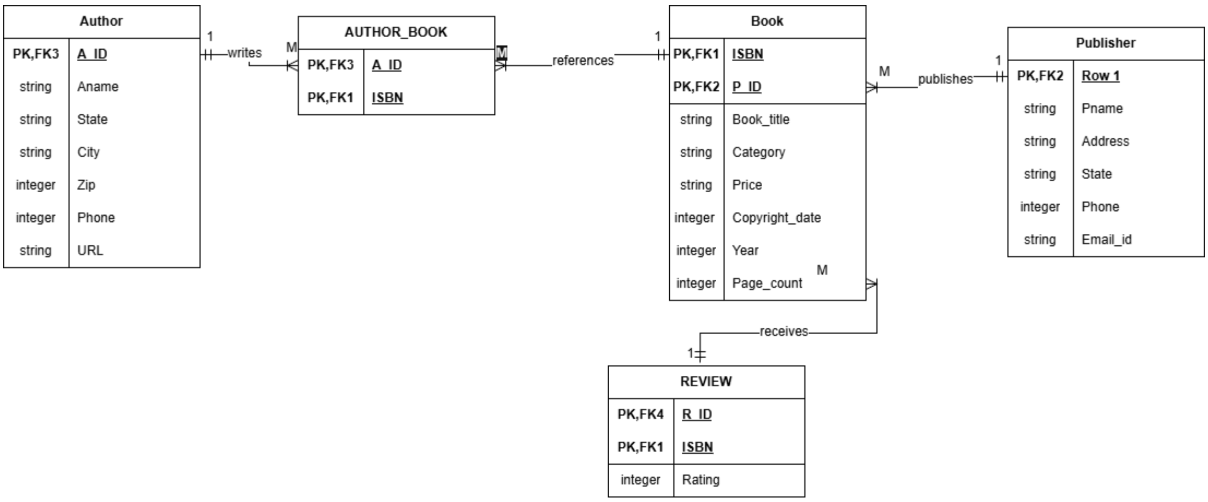
The table has been separated with each entity receiving its respective table.

(Kartik, 2026)

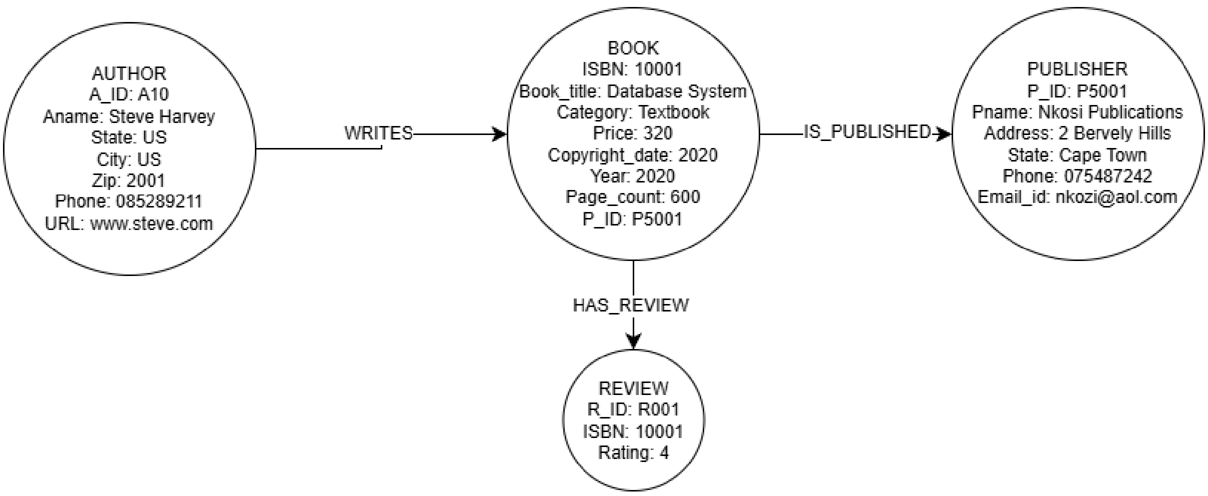
Question 3

3.1

a



b



3.2

1.

$$\pi_{City, Phone, URL}(\sigma_{Aname = "Obaro Adewale"})\ (BOOK)$$

2.

$$\pi_{Pname, Phone, Address}(\sigma_{State = "US"})\ (PUBLISHER)$$

3.

$$\pi_{Book\_title, Price}(\sigma_{Category = "Textbook \wedge Page\_count>400})\ (BOOK)$$

4.

$$\pi_{ISBN, Book\_title, Price}(\sigma_{Category = "Textbook" \vee Category = "Novel"})\ (BOOK)$$

5.

$$\pi_{P\_ID, Pname, Address, Phone}\left(\sigma_{Category = "Novel"}\ (BOOK) \bowtie_{BOOK.P\_ID = PUBLISHER.P\_ID} (PUBLISHER)\right)$$

6.

$$\pi_{Book\_title, Price}\left(\sigma_{Pname = "Nkosi Publications"}\ (PUBLISHER) \bowtie_{PUBLISHER.P\_ID = BOOK.P\_ID} (BOOK)\right)$$

7.

$$\pi_{Book\_title, R\_ID, Rating}\left(\sigma_{Category = "Textbook"}\ (BOOK) \bowtie_{BOOK.ISBN = REVIEW.ISBN} (REVIEW)\right)$$

8.

$$\pi_{Book\_title, Category, Price}\left(\sigma_{Aname = "Steve Harvey"}\ (AUTHOR) \bowtie (AUTHOR\_BOOK) \bowtie (BOOK)\right)$$

(tiefengeist, 2019)

3.3

a)

i)

T1	T2
read(Salary)	
NetSalary := Salary - Vat	
write(NetSalary)	
read(HireDate)	
Age := HireDate - DOB	
write(Age)	

	read(Salary)
	SalaryIncrease := Salary*1.02
	Vat := SlaryIncrease*0.15
	write(SalaryIncrease)
	write(Vat)

ii)

T1	T2
read(Salary)	
	read(Salary)
	SalaryIncrease := Salary*1.02
	Vat := SlaryIncrease*0.15
	write(SalaryIncrease)
	write(Vat)
NetSalary := Salary - Vat	
write(NetSalary)	
read(HireDate)	
Age := HireDate - DOB	
write(Age)	

(pritampaul2k17, 2026)

(*Calculation of Serial and Non-Serial Schedules in DBMS*, 2025)

b)

Query execution time – The response time is the time taken for the database to execute and display a query. You can check this using a SQL Server Profiler.

Resource utilization – The number of resources used when executing a query.

(IBM Informix, 2021)

c)

- Kill the transaction that started the deadlock detection, this is usually the newer one.
- Abort the transaction that has done the least work.

(Arpit Bhayani, 2025)

(Smitha Dinesh Semwal, 2025)

Bibliography

Arpit Bhayani. (2025, March 12). *Why Do Database Deadlock and How Do They Resolve It*. Arpitbhayani.Me. https://arpitbhayani.me/blogs/database-deadlocks/

Calculation of Serial and Non-Serial Schedules in DBMS. (2025, July 28). Geeksforgeeks. https://www.geeksforgeeks.org/dbms/calculation-of-serial-and-non-serial-schedules-in-dbms/

Coursear Staff. (2026, February 21). *What Is Conceptual Data Modeling?* Coursera. https://www.coursera.org/articles/conceptual-data-modeling

IBM Informix. (2021, March 3). *Resource utilization*. IBM Informix. https://www.ibm.com/docs/en/informix-servers/12.10.0?topic=basics-resource-utilization-performance

Ivan Belcic, & Cole Stryker. (n.d.). *What is an entity relationship diagram*. IBM. Retrieved March 13, 2026, from https://www.ibm.com/think/topics/entity-relationship-diagram

Kartik. (2026, February 11). *Introduction to Database Normalization*. Geeksforgeeks. https://www.geeksforgeeks.org/dbms/introduction-of-database-normalization/

pritampaul2k17. (2026, March 12). *Serial vs Non-Serial Schedules in DBMS*. SCRIBD. https://www.scribd.com/document/861572604/Schedules-in-DBMS

Redgate Data Modeler. (2020, December 3). *What Is a Many-to-Many Relationship in a Database? An Explanation with Three Examples*. Redgate. https://www.red-gate.com/blog/many-to-many-relationship/

Smitha Dinesh Semwal. (2025, July 30). *Deadlock in DBMS*. Geeksforgeeks. https://www.geeksforgeeks.org/dbms/deadlock-in-dbms/

tiefengeist. (2019, February 20). *Relational Algebra: the Underpinnings of SQL*. Medium. https://medium.com/@sahirnambiar/relational-algebra-the-underpinnings-of-sql-74959481231a

What is a logical data model? (n.d.). TIBCO. Retrieved March 13, 2026, from https://www.tibco.com/glossary/what-is-a-logical-data-model